

## **Influencing the Synapse**

**Note:** In OneNote, set this worksheet as background. Use "Export" in the simulation and insert the image in the marked space.

At a chemical synapse, information is transmitted with the help of neurotransmitters. These messenger substances are released from the terminal button, pass through the synaptic cleft and bind to receptors on the next cell. This triggers a new signal. Certain substances can change this sequence. Some substances strengthen the effect of neurotransmitters, others block receptors or prevent the breakdown of the messenger substances. As a result, signals become too strong, too weak or are no longer transmitted at all. This can lead to rigid or flaccid paralysis. The chemical synapse is therefore a sensitive site of information transmission. Even small changes can influence the function of neurons and thus the body's reaction.

### **Task 1: Compare sites of action**

Choose two substances from the "Influencing the Synapse" mode. Export two suitable states and insert them below.

Substance 1: \_\_\_\_\_ Substance 2: \_\_\_\_\_

*Insert here*

*Insert here*

**Task 2: Effect on the postsynaptic reaction**

Describe how the selected active substances affect the postsynaptic reaction. Explain whether signal transmission is strengthened, weakened, prolonged or prevented.

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**Task 3: Consequences for the organism**

In the table, add whether the effect of the different substances leads to flaccid paralysis or rigid paralysis. Use the modes of action in the simulation.

| Active substance                | Effect |
|---------------------------------|--------|
| Curare                          |        |
| Botulinumtoxin ( <i>Botox</i> ) |        |
| Sarin                           |        |
| Nikotin                         |        |
| $\alpha$ -Latrotoxin            |        |
| Conotoxin                       |        |